

Derivation of the Probability Density Function for the Weibull Distribution

To derive the Probability Density Function (PDF) for the Weibull Distribution, we start with the cumulative distribution function (CDF) of the exponential distribution, representing the sum of probabilities over a time to failure or time between events, denoted as x :

$$F_X(x; \lambda) = 1 - e^{-\lambda x}; x > 0.$$

If X represents a "time-to-failure," we consider a distribution where the failure rate is proportional to a power of time, parameterized by k [The transformation of the random variable X as $X^k = X \cdot X \dots$ in the cumulative distribution function (CDF) enables the distribution to capture different *accumulated failure* behaviors by adjusting the shape parameter k , thereby influencing the tail behavior (heavy tail) and hazard rate dynamics.]:

$$F_X(x; \lambda, k) = 1 - e^{-\lambda x^k}; x > 0.$$

Here, λ is the scale parameter, indicating the average rate of failure occurrences per unit time, and the shape parameter, k , holds the following interpretations:

- $k < 1$ implies a decreasing failure rate over time.
- $k = 1$ indicates a constant failure rate over time.
- $k > 1$ suggests an increasing failure rate with time.

To determine the PDF, that is the PDF of the Weibull distribution, we differentiate the last CDF with respect to x :

$$\begin{aligned} f(x; \lambda, k) &= \frac{d(F_X(x; \lambda, k))}{dx} = \frac{d}{dx} \left(1 - e^{-\lambda x^k} \right) = 0 - \frac{d}{dx} \left(e^{-\lambda x^k} \right) \\ &= - \frac{d(e^{-\lambda x^k})}{dx} \cdot \frac{dx^k}{dx} = -(-\lambda) \cdot e^{-\lambda x^k} \cdot k \cdot x^{k-1} = \lambda k x^{k-1} e^{-\lambda x^k} \end{aligned}$$

That is, $f(x; \lambda, k) = \lambda k x^{k-1} e^{-\lambda x^k}$. This yields the probability density function (PDF) of the Weibull distribution with parameters λ and k .

The Weibull distribution is commonly used to model the time until a happening occurs, such as the time until a mechanical component fails, the time until a system breaks down, or the time until a product reaches a certain level of deterioration. It is particularly useful in reliability engineering and survival analysis.

The shape of the Weibull distribution can be adjusted using its parameters to capture different failure rate behaviors over time, making it a versatile tool for modeling various real-world phenomena.